

- 1 A toy store gives each of the toys it sells a unique code. The codes are 3-digit hexadecimal codes, for example A4B.

A computer is used to store a database of the codes. The computer has 12-bit registers.

- (a) Give the largest hexadecimal number that can be stored in a 12-bit register.

..... [1]

- (b) Give the largest denary number that can be stored in a 12-bit register.

..... [1]

- (c) Two toys have the hexadecimal codes 429 and 1A3.

Convert the **two** hexadecimal codes to 12-bit binary numbers.

429

1A3 [2]

Working space

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- (d) Two binary numbers that are stored in the registers are 100010100001 and 011100001011.

Convert the **two** binary numbers to hexadecimal numbers.

100010100001

011100001011 [2]

Working space

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